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A New Approach to Field-Oriented Control That Substantially Improves the Efficiency of an Induction Motor with Speed Control

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Abstract: Accurate estimation of the rotor flux angle remains a significant challenge when conventional direct or indirect field-oriented control (FOC) strategies are applied to induction motor drives. This paper proposes a novel method for determining the rotor flux angle under steady-state conditions using only stator voltage and current measurements. An adjustable steady-state detection mechanism is introduced and integrated into a phase-locked loop (PLL)-based indirect field-oriented framework to enable a smooth injection of the actual rotor flux angle into the control system. Both simulation and experimental results validate the effectiveness of the proposed method, demonstrating a significant reduction in stator current compared to conventional FOC approaches under identical load torque conditions.

Keywords: field-oriented control (FOC); induction motors; rotor flux angle estimation; stator current control; phase-locked loop (PLL)

1. Introduction

Induction motors are widely employed in industrial applications due to their inherent ruggedness, high reliability, and cost-effectiveness. When precise and efficient rotor speed control is required, field-oriented control (FOC), also known as vector control, is commonly adopted. With advancements in power electronics and the widespread availability of high-speed digital signal processors (DSPs), model-based control strategies have become the mainstream approach in motor drive systems. In addition, comprehensive training platforms for implementing FOC have also been developed [1,2].

The core concept of FOC lies in transforming the motor dynamic equations into a rotating reference frame in which the d-axis is aligned with the rotor flux vector [3]. Under this condition, rotor dynamics can be described by Equation $L_r \frac{d}{dt} \lambda_{dr}^{rf} + r_r \lambda_{dr}^{rf} = r_r L_m i_{ds}^{rf}$, $T_e = \frac{3PL_m}{4L_r} \lambda_{dr}^{rf} i_{qs}^{rf}$. Accurate control of the current components i_{ds}^{rf} and i_{qs}^{rf} in this frame enables the generation of the desired electromagnetic torque T_e .

Most FOC-based speed control schemes for induction motors share a similar structural configuration, as illustrated in Figure 1. It is important to emphasize that an induction machine's rotor magnetic pole λ_{qdr}^s arises from the relative motion between the stator's rotating magnetic field λ_{qds}^s and the physical rotor. Perfect alignment is not always guaranteed even under steady-state conditions, where λ_{qdr}^s and λ_{qds}^s rotate synchronously. Since the magnitude (λ_{rf}) and direction (θ_{rf}^s) of the rotor flux cannot be directly measured, it is necessary to estimate the rotor flux angle (θ_{rf}^{s*}) prior to applying the control strategy shown in Figure 1. Consequently, various rotor flux angle estimation techniques have been developed, which are generally categorized into two main approaches: direct field-oriented control (DFOC) and indirect field-oriented control (IFOC).



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assumption of (1)—that the back electromotive force (EMF) dominates the stator voltage—becomes invalid due to the reduced influence of EMF at low speeds.

A convergence analysis of the extended Kalman filter (EKF) applied to sensorless control of induction motors was presented in [6]. Although EKF-based methods are capable of estimating rotor flux and speed over a wide speed range, the time-varying nature of the linearized system model introduces additional complexity and may compromise estimation stability. In another study, Chen et al. [5] employed an adaptive full-order observer for rotor flux and speed estimation in sensorless induction motor drives. However, this approach exhibits significant sensitivity to parameter variations, particularly under low-speed regenerative operating conditions.

In [7], Rivera developed two discrete-time models for induction motors using variational integrators, and validated the models through both simulation and experimental results. Subsequently, a complete system framework incorporating both a controller and an observer was constructed based on a sliding mode control strategy, as described in [8]. Although the effectiveness of the proposed system was demonstrated, this study did not address the method's computational efficiency or associated implementation costs.

In the context of direct field-oriented control (FOC), Comanescu et al. [9] proposed incorporating a phase-locked loop (PLL) into the voltage model to estimate the rotor speed. Based on the estimated speed, the cutoff frequency of the low-pass filter (LPF) is adaptively adjusted to enable accurate tracking of the rotor flux angle at medium and high speeds.

In [10], Zhen argued that compared to the previously proposed low-pass filter (LPF), the second-order generalized integrator-based frequency-locked loop (SOGI-FLL), derived from the phase-locked loop (PLL), can improve estimation accuracy, reduce system complexity, and attenuate high-order harmonics in the back electromotive force (EMF). However, its performance degrades significantly during back-EMF transients, resulting in inaccurate frequency estimation and the occurrence of second harmonic oscillations. To address this issue, a dual-axis SOGI-FLL (DSOGI-FLL), constructed using the α and β components, was proposed. This approach not only suppresses the second harmonic oscillations but also enhances estimation accuracy. Nevertheless, similar to the conventional SOGI-FLL, the DSOGI-FLL exhibits poor performance under low-speed conditions.

In indirect field-oriented control (IFOC), the slip frequency ($S\omega_e$) is first estimated, and the rotor flux angle is then derived using

$$\theta_{rf}^s = \int_0^t S\omega_e(\tau) + \omega_r(\tau) d\tau \quad (2)$$

where $\omega_r = \omega_m \frac{P}{2}$ represents the product of the actual rotor speed and the number of pole pairs. Strictly speaking, (2) is valid only when the slip frequency is defined as the difference between the rotor flux angular velocity and the rotor mechanical angular velocity ($S\omega_e \equiv \omega_{rf} - \omega_r$) [11]. Under steady-state conditions, this value approximates the difference between the synchronous speed and the rotor electrical speed ($\omega_e - \omega_r$).

Figure 3a illustrates the representative structure of a typical indirect field-oriented control (IFOC) scheme [3]. In [12], the concept of multiple three-phase systems was further extended to enable the control of multiphase induction motors.

Moreover, the concept of slip frequency has been applied to widely used and cost-effective induction machines to improve the inefficiencies associated with conventional control strategies [13]. In traditional approaches, the slip frequency is estimated by determining the d-axis and q-axis current references (i_{ds}^{rf*} and i_{qs}^{rf*}) based on the torque and flux commands, respectively. An estimate of the slip frequency can then be obtained, as illustrated in Figure 3b.

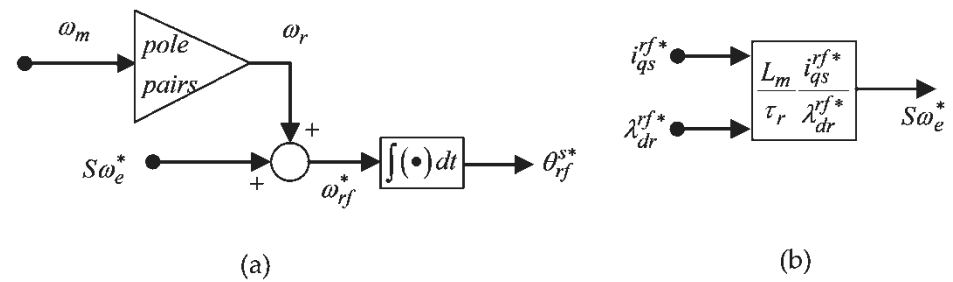


Figure 3. (a) The common parts of all indirect FOC structures. (b) The most conventional indirect FOC method of estimating the slip frequency.

However, this method does not account for transient behavior, which has led to the development of several improved approaches aimed at enhancing performance under dynamic operating conditions [14,15].

For example, Rubin [14] analyzed an equivalent steady-state circuit of the induction motor, as illustrated in Figure 4, and derived the following expression:

$$\rho \triangleq \frac{i_{ds}^p}{i_{qs}^p} = \frac{\beta_2(S\omega_e)^2 + \beta_1 S\omega_e + \beta_0}{\alpha_2(S\omega_e)^2 + \alpha_0}. \quad (3)$$

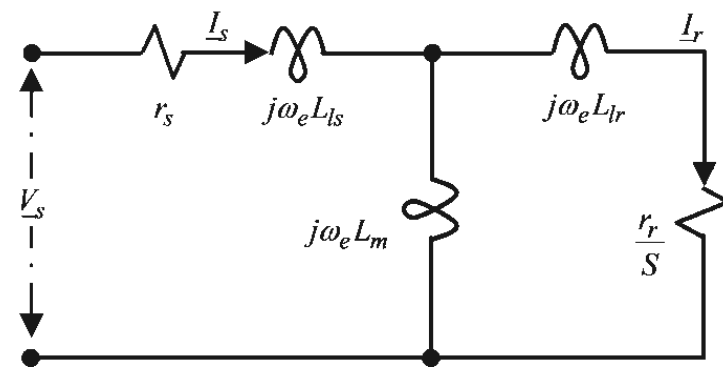


Figure 4. The equivalent circuit of induction motors under steady state.

In this expression, $\alpha_2 = \omega_e L_{lr} (L_m^2 - L_{ls} L_{lr})$, $\alpha_0 = -\omega_e r_r^2 L_{ls}$, $\beta_2 = r_s L_{lr}^2$, $\beta_1 = \omega_e r_r L_m^2$, $\beta_0 = r_s r_r^2$, and the superscript p indicate that the d-axis of the reference frame is aligned with the stator voltage vector. Based on this formulation, the slip frequency is expressed as follows:

$$S\omega_e = \frac{\beta_1 \pm \sqrt{\beta_1^2 - 4(\rho\alpha_0 - \beta_0)(\rho\alpha_2 - \beta_2)}}{2(\rho\alpha_2 - \beta_2)}. \quad (4)$$

Furthermore, using the definition of ρ , along with measured values of the stator current and voltage, the following relationship can be derived:

$$\rho = \frac{3i_{qds}^s T v_{qds}^s - i_{0s} v_{0s}}{\sqrt{3} \mathbf{1}^T (i_{qds}^s T \times v_{qds}^s T)}. \quad (5)$$

Here, i_{0s} and v_{0s} represent the current and voltage at the neutral point, respectively. Equation (3) indicates that ρ is a function of $S\omega_e$, and the power factor reaches its optimal value when the condition

$$\left. \frac{d\rho}{d(S\omega_e)} \right|_{S=S_c} = 0 \Rightarrow \rho_{\max} = f(S\omega_e)|_{S=S_c}$$

is satisfied, where S_c represents the critical slip. Accordingly, this criterion can be used to determine the valid solution of (4).

Another approach, presented in [15] and illustrated by the structure in Figure 5, addresses this issue by feeding the stator current ($|i_{qds}(k)|$) into a fuzzy logic controller. The fuzzy control rules are designed based on the principles depicted in Figure 6. Once the motor reaches steady-state operation, the variation in rotor speed ($|\Delta\omega_r(k)|$) remains below a predefined threshold (ε). The most recent output of the fuzzy controller is then added to the slip frequency estimated by the conventional indirect FOC scheme.

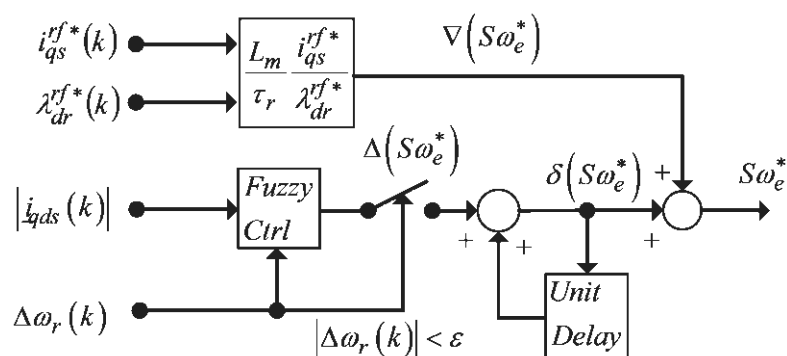


Figure 5. The estimation method used in [15].

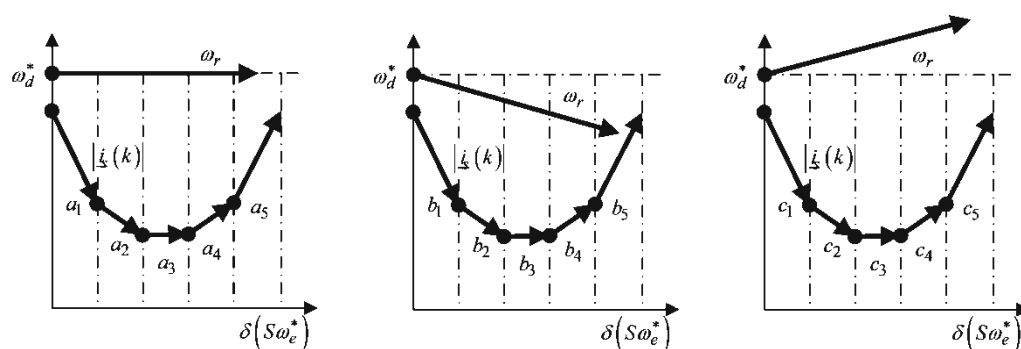


Figure 6. The fuzzy rules used in [4].

The physical interpretation of Figure 6 is as follows: under the condition that the electrical rotor speed (ω_r) remains constant at its steady-state value (ω_d^*), the output parameter $\Delta(S\omega_e^*)$ is adjusted to modify $\delta(S\omega_e^*)$ such that the resulting stator current ($|i_{qds}(k)|$) is minimized. This condition yields the optimal estimation of the slip frequency ($S\omega_e^*$).

However, during this process, the deviation between ω_r and ω_d^* may increase as $\delta(S\omega_e^*)$ grows. In such cases, the system departs from steady-state operation. To restore stability, the variable $\delta(S\omega_e^*)$ must be decreased, regardless of the variation in $|i_{qds}(k)|$, which implies that $\Delta(S\omega_e^*)$ becomes negative.

In summary, the optimal field-oriented control (FOC) under steady-state conditions corresponds to the configuration that minimizes the stator current ($|i_{qds}(k)|$).

Although the procedures described above yield an accurate or optimal slip frequency ($S\omega_e^*$) under steady-state conditions, the estimation of θ_{rf}^{s*} remains affected by accumulated error during transients due to the integration process in (2). As a result, the slip frequency derived purely under steady-state assumptions cannot ensure accurate estimation of the rotor flux angle (θ_{rf}^s).

Therefore, unless the slip frequency ($S\omega_e^*$) can be precisely estimated during transients, the resulting field-oriented control will be inaccurate, limiting the effectiveness of these conventional approaches.

Recently, Milic et al. [16] proposed a rotor speed sensing technique based on the observation that discontinuities in the magnetic path, caused by rotating rotor slots, induce high-frequency harmonics in the stator phase currents. These harmonics are utilized to infer the rotor speed. However, the stator current is also affected by magnetic and electrical interferences, necessitating the use of filters to extract the true speed signal. Since the machine operates over a wide speed range, the filter's cutoff frequency must adapt to the varying electrical frequency. To achieve this, the authors incorporated complex algorithms such as phase-locked loops (PLLs). Nevertheless, skewed slot designs—commonly adopted in motor manufacturing to reduce cogging torque—conflict with the assumptions required by this approach. As a result, the practical viability of this technique remains under investigation.

A combined FOC and Direct Torque Control (DTC) approach was proposed to enhance the performance of three-phase induction motors across varying torque loads. This method utilizes a dynamic reconfiguration between control algorithms to optimize performance during different operating conditions [17]. A sensorless speed control method employing a Model Reference Adaptive System (MRAS) and a deadbeat regulator was introduced to improve affordability and efficiency by eliminating the need for physical speed sensors [18]. A compensated direct field-oriented sensorless control using a Luenberger observer was developed for induction motors connected via long cables and filters, addressing challenges in submarine applications where sensor deployment is impractical [19]. A hybrid sensorless speed estimator based on flux error was presented for electric vehicle applications, aiming to improve speed tracking accuracy in indirect FOC systems [20]. A nonlinear field-oriented control technique utilizing tensor product representations and linear matrix inequalities was proposed to enhance control performance in induction machines [21].

The technique proposed in this paper does not fall under conventional indirect field-oriented control (IFOC) or direct field-oriented control (DFOC). Instead, it integrates the advantages of both approaches while avoiding their respective limitations. As shown in Figure 3b, conventional IFOC estimates the rotor flux angle ($S\omega_e^*$) based on command values (i_{qs}^{rf*} and λ_{dr}^{rf*}), which are then integrated and treated as the actual rotor flux angle to be fed back into the control system. Clearly, such an approach may deviate from real-world behavior. For example, although Ozdemir [22] incorporated a phase-locked loop (PLL) algorithm along with an adaptive model to mitigate issues associated with pure integration, the model still relies on the structural framework depicted in Figure 3b.

Similarly, the sensorless DFOC scheme shown in Figure 2b continues to depend on integration to derive λ_{qds}^s , which is then fed back to compute λ_{qdr}^s and eventually obtain θ_{rf}^{s*} . However, continuous integration inevitably accumulates error without opportunities for correction. Therefore, the objectives of this paper are as follows:

- Pure Integration-Free and PLL-Based Estimation: Eliminates numerical drift and error accumulation typical in conventional indirect FOC.
- Hybrid Control Strategy: Effectively addresses transient performance without sacrificing steady-state accuracy.
- Experimentally Validated Efficiency: Demonstrates significant real-world energy savings and efficiency improvements.
- Moderate Complexity and Cost: Offers practical implementation advantages suitable for industry-standard DSP platforms.

2. New Field-Oriented Control (FOC) Technique

Therefore, the objective of this paper is to accurately estimate the rotor flux angle (θ_{rf}^{s*}) under steady-state conditions using only directly measurable stator current and voltage signals, while avoiding integration operations and adopting a continuously correctable approach. This

method addresses the limitations of conventional FOC schemes, particularly the estimation inaccuracies that become more pronounced at low speeds. The estimated rotor flux angle is directly incorporated into the FOC algorithm to enhance real-time performance.

According to the extended formulation presented by Novotny [3], the control equation of the stator circuit for a squirrel cage induction motor, expressed in the rotating d–q reference frame with angular velocity ω , is given by

$$\begin{aligned} v_{ds}^{\omega} &= r_s i_{ds}^{\omega} + D\lambda_{ds}^{\omega} - \omega \lambda_{qs}^{\omega}, \lambda_{ds}^{\omega} = L_s i_{ds}^{\omega} + L_m i_{dr}^{\omega} \\ v_{qs}^{\omega} &= r_s i_{qs}^{\omega} + D\lambda_{qs}^{\omega} + \omega \lambda_{ds}^{\omega}, \lambda_{qs}^{\omega} = L_s i_{qs}^{\omega} + L_m i_{qr}^{\omega} \end{aligned} \quad (6)$$

where r_s and L_s denote the stator resistance and inductance, respectively. The corresponding rotor circuit equation is expressed in

$$\begin{aligned} v_{dr}^{\omega} &= r_r i_{dr}^{\omega} + D\lambda_{dr}^{\omega} - (\omega - \omega_r) \lambda_{qr}^{\omega} = 0, \lambda_{dr}^{\omega} = L_r i_{dr}^{\omega} + L_m i_{ds}^{\omega} \\ v_{qr}^{\omega} &= r_r i_{qr}^{\omega} + D\lambda_{qr}^{\omega} + (\omega - \omega_r) \lambda_{dr}^{\omega} = 0, \lambda_{qr}^{\omega} = L_r i_{qr}^{\omega} + L_m i_{qs}^{\omega} \end{aligned} \quad (7)$$

where $D \equiv \frac{d}{dt}$ represents the time derivative operator.

Substituting the magnetic flux expressions from (6) and (7) into the voltage equations yields

$$\begin{aligned} v_{qs}^{\omega} &= \omega L_s i_{ds}^{\omega} + (r_s + L_s D) i_{qs}^{\omega} + \omega L_m i_{dr}^{\omega} + L_m D i_{qr}^{\omega} \\ v_{ds}^{\omega} &= (r_s + L_s D) i_{ds}^{\omega} - \omega L_s i_{qs}^{\omega} + L_m D i_{dr}^{\omega} - \omega L_m i_{qr}^{\omega} \\ v_{dr}^{\omega} &= L_m D i_{ds}^{\omega} - (\omega - \omega_r) L_m i_{qs}^{\omega} + (r_r + L_r D) i_{dr}^{\omega} - (\omega - \omega_r) L_r i_{qr}^{\omega} = 0 \\ v_{qr}^{\omega} &= (\omega - \omega_r) L_m i_{ds}^{\omega} + L_m D i_{qs}^{\omega} + (\omega - \omega_r) L_r i_{dr}^{\omega} + (r_r + L_r D) i_{qr}^{\omega} = 0 \end{aligned} \quad (8)$$

$\lambda_{qr}^{rf} \equiv 0 \Rightarrow D\lambda_{qr}^{rf} = 0$ is obtained under the assumption that the d-axis of the rotating reference frame is continuously aligned with the rotor flux vector. Under this alignment condition, the original stator voltage Equation (6) simplifies to the following set of reduced equations:

$$0 = r_r i_{dr}^{rf} + D\lambda_{dr}^{rf}, \quad (9)$$

$$0 = r_r i_{qr}^{rf} + (\omega_{rf} - \omega_r) \lambda_{dr}^{rf}, \quad (10)$$

$$\lambda_{dr}^{rf} = L_r i_{dr}^{rf} + L_m i_{ds}^{rf}, \quad (11)$$

$$0 = L_r i_{qr}^{rf} + L_m i_{qs}^{rf}. \quad (12)$$

From Equation (11), an intermediate variable $i_{dr}^{rf} = \frac{\lambda_{dr}^{rf} - L_m i_{ds}^{rf}}{L_r}$ is derived and subsequently substituted into Equation (9), yielding the following:

$$(1 + \tau_r D) \lambda_{dr}^{rf} = L_m i_{ds}^{rf} \Rightarrow \lambda_{dr}^{rf} = \frac{L_m}{1 + \tau_r D} i_{ds}^{rf}. \quad (13)$$

In (13), $\tau_r \triangleq L_r / r_r$ denotes the rotor time constant, and s represents the Laplace operator. It is important to note that (13) holds only under relaxed motor conditions (i.e., steady-state operation). Additionally, Equation (12) implies the following relationship:

$$i_{qr}^{rf} = -\frac{L_m}{L_r} i_{qs}^{rf}. \quad (14)$$

By combining (10), (13), and (14), the following expression is derived: $S\omega_e(s) \triangleq \omega_{rf}(s) - \omega_r(s) = \frac{1 + \tau_r s}{\tau_r} \frac{i_{qs}^{rf}(s)}{i_{ds}^{rf}(s)} \Rightarrow \frac{i_{qs}^{rf}(s)}{i_{ds}^{rf}(s)} = \frac{\tau_r}{1 + \tau_r s} S\omega_e(s)$. Substituting the final value theorem into the above yields the following:

$$\lim_{t \rightarrow \infty} \frac{i_{qs}^{rf}(t)}{i_{ds}^{rf}(t)} = \frac{\lim_{t \rightarrow \infty} i_{qs}^{rf}(t)}{\lim_{t \rightarrow \infty} i_{ds}^{rf}(t)} = \frac{\lim_{s \rightarrow 0} i_{qs}^{rf}(s)}{\lim_{s \rightarrow 0} i_{ds}^{rf}(s)} = \lim_{s \rightarrow 0} \frac{i_{qs}^{rf}(s)}{i_{ds}^{rf}(s)} = \lim_{s \rightarrow 0} \frac{\tau_r}{1 + \tau_r s} S\omega_e(s) = \tau_r \lim_{s \rightarrow 0} S\omega_e(s) \equiv \tau_r \cdot y. \quad (15)$$

Based on the geometric relationship illustrated in Figure 7, the following equation is obtained:

$$\lim_{t \rightarrow \infty} \frac{i_{qs}^{rf}(t)}{i_{ds}^{rf}(t)} = \lim_{t \rightarrow \infty} \frac{i_{qds}(t) \sin \gamma(t)}{i_{qds}(t) \cos \gamma(t)} = \lim_{t \rightarrow \infty} \tan \gamma(t) = \tan \left[\lim_{t \rightarrow \infty} \gamma(t) \right] \Rightarrow \lim_{t \rightarrow \infty} \gamma(t) = \tan^{-1}(\tau_r \cdot y). \quad (16)$$

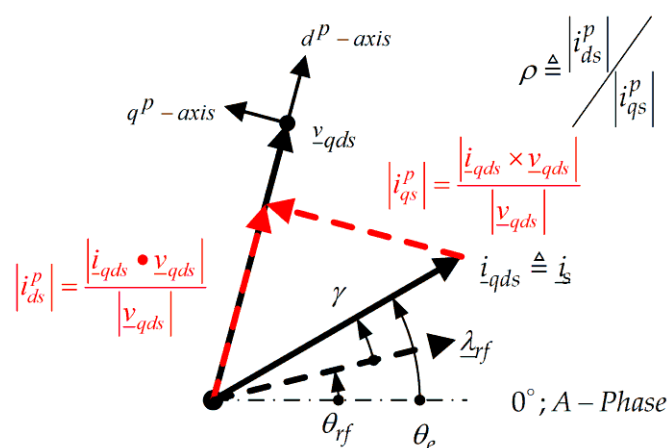


Figure 7. The geometric meaning of ρ .

Here, $\gamma \triangleq \theta_e - \theta_{rf}$ represents the angle by which the stator current vector i_{qds} leads the rotor flux vector $\underline{\lambda}_{qdr}$. If the d-axis of the reference frame is aligned with the stator voltage vector ($\underline{v}_{qds}(t)$), the variable $v_{qs}^p \equiv 0$ can be obtained, and Equation (8) simplifies into the following two expressions:

$$\begin{aligned} v_{ds}^p &= (r_s + L_s D) i_{ds}^p - \omega_e L_s i_{qs}^p + L_m D i_{dr}^p - \omega_e L_m i_{qr}^p \\ 0 &= \omega_e L_s i_{ds}^p + (r_s + L_s D) i_{qs}^p + \omega_e L_m i_{dr}^p + L_m D i_{qr}^p \end{aligned} \quad (17)$$

$$\begin{aligned} 0 &= L_m D i_{ds}^p - (\omega_e - \omega_r) L_m i_{qs}^p + (r_r + L_r D) i_{dr}^p - (\omega_e - \omega_r) L_r i_{qr}^p \\ 0 &= (\omega_e - \omega_r) L_m i_{ds}^p + L_m D i_{qs}^p + (\omega_e - \omega_r) L_r i_{dr}^p + (r_r + L_r D) i_{qr}^p \end{aligned} \quad (18)$$

Applying the Laplace transform to both equations, the solution to (17) yields the intermediate terms $i_{dr}^p(s) = \frac{sv_{ds}^p - (\omega_e^2 L_s + r_s s + L_s s^2) i_{ds}^p - \omega_e r_s i_{qs}^p}{L_m (\omega_e^2 + s^2)}$ and $i_{qr}^p(s) = \frac{\omega_e r_s i_{ds}^p - (\omega_e^2 L_s + r_s s + L_s s^2) i_{qs}^p - \omega_e v_{ds}^p}{L_m (\omega_e^2 + s^2)}$.

These results are then substituted into (18) to derive the following:

$$a_2 i_{d_s}^p - a_3 i_{q_s}^p = a_1 v_{d_s}^p, \quad -b_2 i_{d_s}^p - b_3 i_{q_s}^p = b_1 v_{d_s}^p, \quad (19)$$

where $a_1 = \frac{1}{L_m(\omega_e^2 + s^2)}[(\omega_e - \omega_r)L_r\omega_e + (r_r + L_rs)s]$, $a_2 = \frac{1}{L_m(\omega_e^2 + s^2)}[(r_r + L_rs)(\omega_e^2 L_s + r_s s + L_s s^2) + \omega_e(\omega_e - \omega_r)r_s L_r - L_m^2 s(\omega_e^2 + s^2)]$, $a_3 = \frac{1}{L_m(\omega_e^2 + s^2)}[(\omega_e - \omega_r)L_r(\omega_e^2 L_s + r_s s + L_s s^2) - \omega_e r_s(r_r + L_rs) - (\omega_e - \omega_r)L_m^2(\omega_e^2 + s^2)]$, and $b_1 = \frac{1}{L_m(\omega_e^2 + s^2)}[(\omega_e - \omega_r)L_r s - (r_r + L_rs)\omega_e]$, $b_2 = -a_3$, $b_3 = -a_2$ are system-dependent coefficients.

By substituting the previously obtained coefficients into Equation (19), the closed-form solution is derived as follows:

$$\frac{i_{ds}^p(s)}{i_{qs}^p(s)} = \frac{\widehat{a}_3 s^3 + \widehat{a}_2 s^2 + \widehat{a}_1 s + \widehat{a}_0}{\widehat{b}_2 s^2 + \widehat{b}_1 s + \widehat{b}_0} \quad (20)$$

In this expression, $\widehat{a}_3 = -L_r(L_m^2 - L_r L_s)$, $\widehat{a}_2 = L_r(r_s L_r + r_r L_s) - r_r(L_m^2 - L_r L_s)$, $\widehat{a}_1 = r_r(2r_s L_r + r_r L_s) - (\omega_e - \omega_r)^2 L_r(L_m^2 - L_r L_s)$, $\widehat{a}_0 = r_r^2 r_s + r_s L_r^2 (\omega_e - \omega_r)^2 + \omega_e r_r L_m^2 (\omega_e - \omega_r)$, $\widehat{b}_2 = \omega_e L_r(L_m^2 - L_r L_s)$, $\widehat{b}_1 = r_r[\omega_e(L_m^2 - 2L_r L_s) + (\omega_e - \omega_r)L_m^2]$, and $\widehat{b}_0 = (\omega_e - \omega_r)^2 \omega_e L_r(L_m^2 - L_r L_s) - r_r^2 L_s \omega_e$ represent system-specific constants and intermediate variables derived from the stator and rotor parameters.

Under steady-state conditions, the following relation holds: $\omega_{rf} \rightarrow \omega_e \Rightarrow S\omega_e \rightarrow (\omega_e - \omega_r)$. Furthermore, based on the definition of ρ and its geometric interpretation illustrated in Figure 7, the following expression can be derived [14]:

$$\rho \equiv \frac{i_{ds}^p(t)}{i_{qs}^p(t)} = \frac{i_{qds} \bullet v_{qds}}{i_{qds} \times v_{qds}} = \frac{i_{ds}^s v_{ds}^s + i_{qs}^s v_{qs}^s}{i_{qs}^s v_{ds}^s - i_{ds}^s v_{qs}^s} \quad (21)$$

Similarly, applying the final value theorem to (20) yields the following expression:

$$\lim_{t \rightarrow \infty} \rho = \lim_{t \rightarrow \infty} \frac{i_{ds}^p(t)}{i_{qs}^p(t)} = \frac{\lim_{t \rightarrow \infty} i_{ds}^p(t)}{\lim_{t \rightarrow \infty} i_{qs}^p(t)} = \frac{\lim_{s \rightarrow 0} s i_{ds}^p(s)}{\lim_{s \rightarrow 0} s i_{qs}^p(s)} = \lim_{s \rightarrow 0} \frac{i_{ds}^p(s)}{i_{qs}^p(s)} = \lim_{s \rightarrow 0} \frac{r_s L_r^2 (S\omega_e)^2 + \omega_e r_r L_m^2 (S\omega_e) + r_r^2 r_s}{\omega_e L_r (L_m^2 - L_r L_s) (S\omega_e)^2 - \omega_e r_r^2 L_s} = \frac{r_s L_r^2 y^2 + \omega_e r_r L_m^2 y + r_r^2 r_s}{\omega_e L_r (L_m^2 - L_r L_s) y^2 - \omega_e r_r^2 L_s}.$$

Let us define $\alpha_2 = \omega_e L_r(L_m^2 - L_r L_s)$, $\alpha_0 = -\omega_e r_r^2 L_s$, $\beta_2 = -r_s L_r^2$, $\beta_1 = -\omega_e r_r L_m^2$, $\beta_0 = -r_r^2 r_s$ as auxiliary variables representing [brief description, if needed], then Equation (22) can be obtained: $\left[\left(\lim_{t \rightarrow \infty} \rho \right) \alpha_2 + \beta_2 \right] y^2 + \beta_1 y + \left[\left(\lim_{t \rightarrow \infty} \rho \right) \alpha_0 + \beta_0 \right] = 0$

$$\Rightarrow y = \frac{-\beta_1 \pm \sqrt{\beta_1^2 - 4 \left[\left(\lim_{t \rightarrow \infty} \rho \right) \alpha_2 + \beta_2 \right] \left[\left(\lim_{t \rightarrow \infty} \rho \right) \alpha_0 + \beta_0 \right]}}{2 \left[\left(\lim_{t \rightarrow \infty} \rho \right) \alpha_2 + \beta_2 \right]} \quad (22)$$

As shown in the following expression, the steady-state value of ρ can be derived from the components of the voltage vector v_{qds} (output of the controller) and the feedback current i_{qds} (measurable) in the stationary reference frame, denoted by the superscript s . This value is then applied to (21) to determine y , and subsequently used in (16) to obtain the steady-state angle $\gamma(t) = \theta_e - \theta_{rf}$, which represents the phase shift between the stator current and the rotor flux vector. In addition, the absolute angle $\theta_{is}^s(t)$ of the stator current vector $i_{qds}^s(t)$ can be directly measured from the phase difference between the actual current waveform and the voltage waveform v_{qds} . Therefore, under steady-state conditions, the rotor flux angle is given by

$$\theta_{rf}^s(t) = \theta_{is}^s(t) - \gamma(t) \quad (23)$$

Based on this analysis, the overall system architecture proposed in this study is illustrated in Figure 8.

Conceptually, the outputs of the $\theta_{is}^s = \tan^{-1} \left(i_{qs}^s / i_{ds}^s \right)$ and $\omega_e^s = \frac{d}{dt}(\theta_{is}^s)$ computation blocks produce intermediate values, which are then fed into the θ_{rf}^s calculator to compute α_2 , α_0 , β_2 , β_1 , and β_0 . These results are subsequently applied to (21), (16), and (23), respectively, to solve for y , $\gamma(t)$, and θ_{rf}^s .

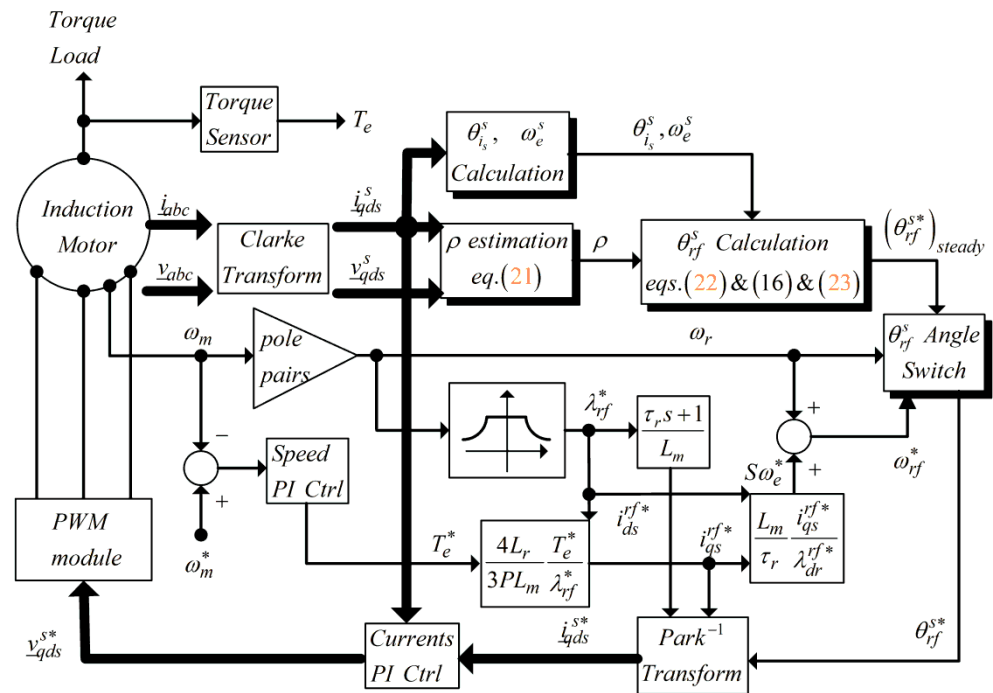


Figure 8. System structure of the proposed method in Stage I.

However, due to the fact that the domain of variable $\tan^{-1}$ is defined over $-\frac{\pi}{2} \sim \frac{\pi}{2}$ with non-differentiable endpoints, the above computational process poses challenges for numerical implementation. This discontinuity hinders the accurate calculation of intermediate variables θ_{is}^s , ω_e^s , and $\gamma(t)$ through direct analytical methods.

To address this issue, the θ_{is}^s and ω_e^s computation blocks in Figure 8 are replaced by a phase-locked loop (PLL) module, as shown in Figure 9. This modification eliminates the dependence on the function involving $\tan^{-1}$, thereby enabling a more robust and continuous estimation process.

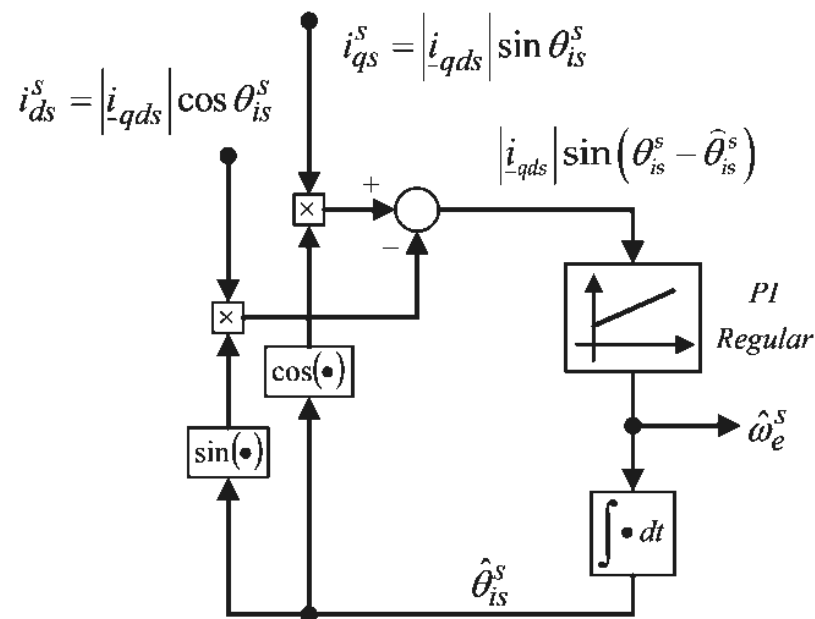


Figure 9. The functional blocks of $\hat{\omega}_e^s$ generator.

The outputs of the ω_e^s and ρ estimators are subsequently fed into (21) and (16) to compute $\tau_r y = \tan[\gamma(t)]$. This result is then input into the computational structure shown in Figure 10. The calculation framework in Figure 10 is based on the expression given in

$$\begin{aligned} A \sin(\theta_{is}^s - \gamma) &= i_{qs}^s \cos \gamma - i_{ds}^s \sin \gamma = A \sin(\theta_{rf}^s) \\ A \cos(\theta_{is}^s - \gamma) &= i_{ds}^s \cos \gamma + i_{qs}^s \sin \gamma = A \cos(\theta_{rf}^s) \end{aligned} \quad (24)$$

which involves variables $i_{qs}^s = A \sin(\theta_{is}^s)$ and $i_{ds}^s = A \cos(\theta_{is}^s)$.

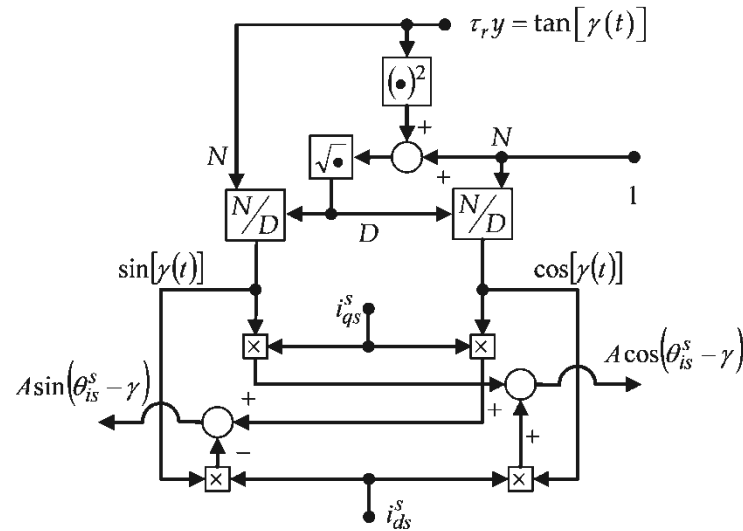


Figure 10. The functional blocks of $\hat{\theta}_{rf}^s$ calculator.

Although (24) reveals information related to the rotor flux angle, it still depends on the estimation of $\hat{\theta}_{rf}^{s*} = \tan^{-1} \left[\frac{\sin(\theta_{is}^s - \gamma)}{\cos(\theta_{is}^s - \gamma)} \right]$. This necessitates the use of phase-locked loop (PLL) techniques once again. To address this, we propose an enhanced method that integrates the PLL concept with conventional indirect FOC, and introduces an angle switching module, as illustrated in Figure 11.

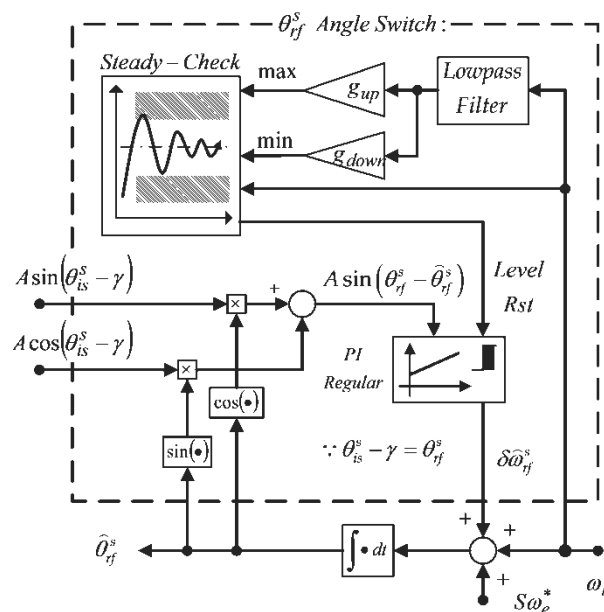


Figure 11. The functional blocks of θ_{rf}^s angle Switch.

Since our derivation is based on the final value theorem, which implies steady-state operation, the structure shown in Figure 11 includes a mechanism to determine whether the system has reached steady state and to coordinate the corresponding transient mode of operation. The actual rotor speed, denoted as ω_r , is obtained from a dedicated speed sensor. This signal is passed through a low-pass filter to extract the low-frequency component. Based on this filtered signal, an adaptive tolerance band is defined by the upper and lower bounds $g_{up} = 1 + \Delta$ and $g_{down} = 1 - \Delta$, where Δ represents a tunable parameter that determines the allowable bandwidth. When the dynamic behavior of ω_r exceeds this predefined range, the system identifies it as a transient state deviation. Consequently, a control signal is generated to reset both the output $\delta\hat{\omega}_{rf}^s$ and the integrator state of the PI regulator to zero, thereby reverting the control scheme back to the conventional Indirect Field-Oriented Control (IFOC) mode.

The proposed module is built upon the conventional indirect FOC structure and incorporates the developed PLL-based framework. Under non-steady-state conditions—such as motor startup, acceleration, deceleration, or during significant torque disturbances—the rotor speed often falls outside the predefined steady-state range. When this occurs, the detection mechanism activates a low-level “Level_Rst” signal, which suppresses the output of the proportional-integral (PI) regulator, effectively holding it at zero. At this stage, the PLL is disabled and unable to operate, and the system reverts to a traditional indirect FOC scheme to estimate the rotor flux angle (θ_{rf}^{s*}), as depicted in Figure 3a,b.

Once the system re-enters the predefined steady-state range, the “Level_Rst” signal returns to a high level, thereby activating the proposed control strategy. The phase-locked loop (PLL) utilizes its output ($\delta\hat{\omega}_{rf}^s$) to adjust the estimated angle ($\hat{\theta}_{rf}^s = \theta_{rf}^{s*}$), ensuring close tracking of the desired angle (θ_{rf}^{s*}). Through this repeated switching mechanism, the system leverages the robustness of the traditional indirect FOC during transients and transitions to a more accurate estimation mode during steady-state. Consequently, the initial architecture shown in Figure 8 is further refined into the second-stage structure illustrated in Figure 12.

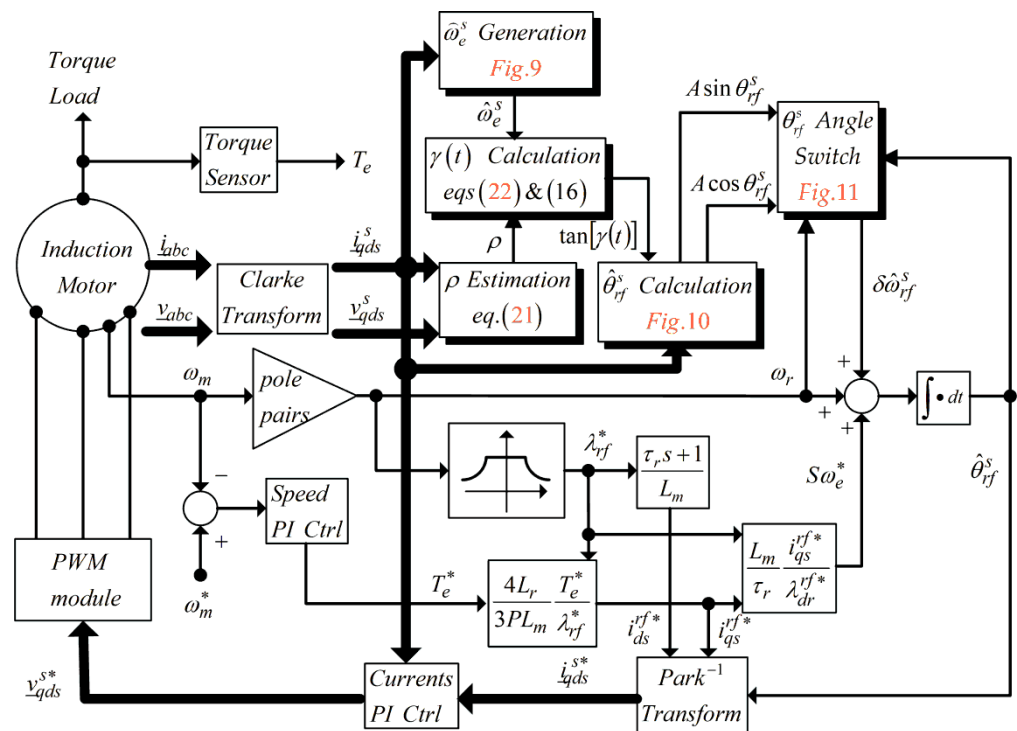


Figure 12. System structure of the proposed method in Stage II.

3. Simulation and Result

The proposed method was verified through simulations conducted using MATLAB/Simulink. An induction motor was modeled using the built-in asynchronous machine module, with its rated values and parameters summarized in Table 1. The overall simulation architecture is illustrated in Figure 12.

Table 1. The ratings and parameters of this modular motor.

Quantity	Rated Value
Power	5 (hp)
Speed	1750 (rpm)
Voltage	460 (V)
Frequency	60 (Hz)
Rotor's inertia	$J = 0.02 (\text{kg} \cdot \text{m}^2)$
Damping friction	$B = 0.005752 \left(\frac{\text{Ns}}{\text{m}} \right) 202$
Number of poles	4
Rotor's resistance	$r_r = 1.083 (\Omega)$
Stator's resistance	$r_s = 1.115 (\Omega)$
Inductance of flux leakage	$L_{ls} = L_{lr} = 0.005974 (\text{H})$
Mutual inductance	$L_m = 0.2037 (\text{H})$

The simulation study compares two control strategies: the proposed field-oriented control (FOC) method and the conventional indirect FOC approach. Two separate simulation scenarios were designed. In the first scenario, only the speed command experiences a step change in $10\pi \rightarrow 100\pi \left(\frac{\text{rad}}{\text{s}} \right)$ at $t = 1$ s. In the second scenario, only the torque load is subjected to a step change in $2.04 \rightarrow 18.36 (\text{Nm})$ at $t = 1$ s. It is important to note that prior to the step disturbances, the system had already reached a steady-state condition under a constant speed command $100\pi \left(\frac{\text{rad}}{\text{s}} \right)$ and a constant load torque $18.36 (\text{Nm})$.

Figure 13 presents the simulation results. In all figures, the green and brown curves represent the performance of the proposed FOC method and the conventional FOC method, respectively. The left-hand side of the figure—Figure 13a–d—corresponds to the scenario in which a step change in the speed command is applied at $t = 1$ s under a fixed load torque.

Specifically, Figure 13a shows the rotor speed response. Figure 13b illustrates the variation in electromagnetic torque. Figure 13c compares the rotor flux angle estimations from both methods. It can be observed that while both methods produce similar results during transient conditions, their differences become increasingly evident once the system reaches steady state. Figure 13d shows the magnitude of the stator current vector. After the step disturbance, the current amplitude under the proposed method gradually decreases, indicating improved efficiency.

The right-hand side of Figure 13e–h—illustrates the scenario in which a step change in load torque is applied at $t = 1$ s while the speed command remains constant.

Figure 13e shows the rotor speed response to the sudden load variation. Figure 13f presents the corresponding electromagnetic torque. Figure 13g compares the estimated rotor flux angles obtained from the two methods. Figure 13h displays the magnitude of the stator current vector. Similar trends to those observed in the speed-command variation case can be seen here: during the transient period, both methods perform comparably; however, once the system stabilizes, the proposed method exhibits superior estimation accuracy and improved current efficiency.

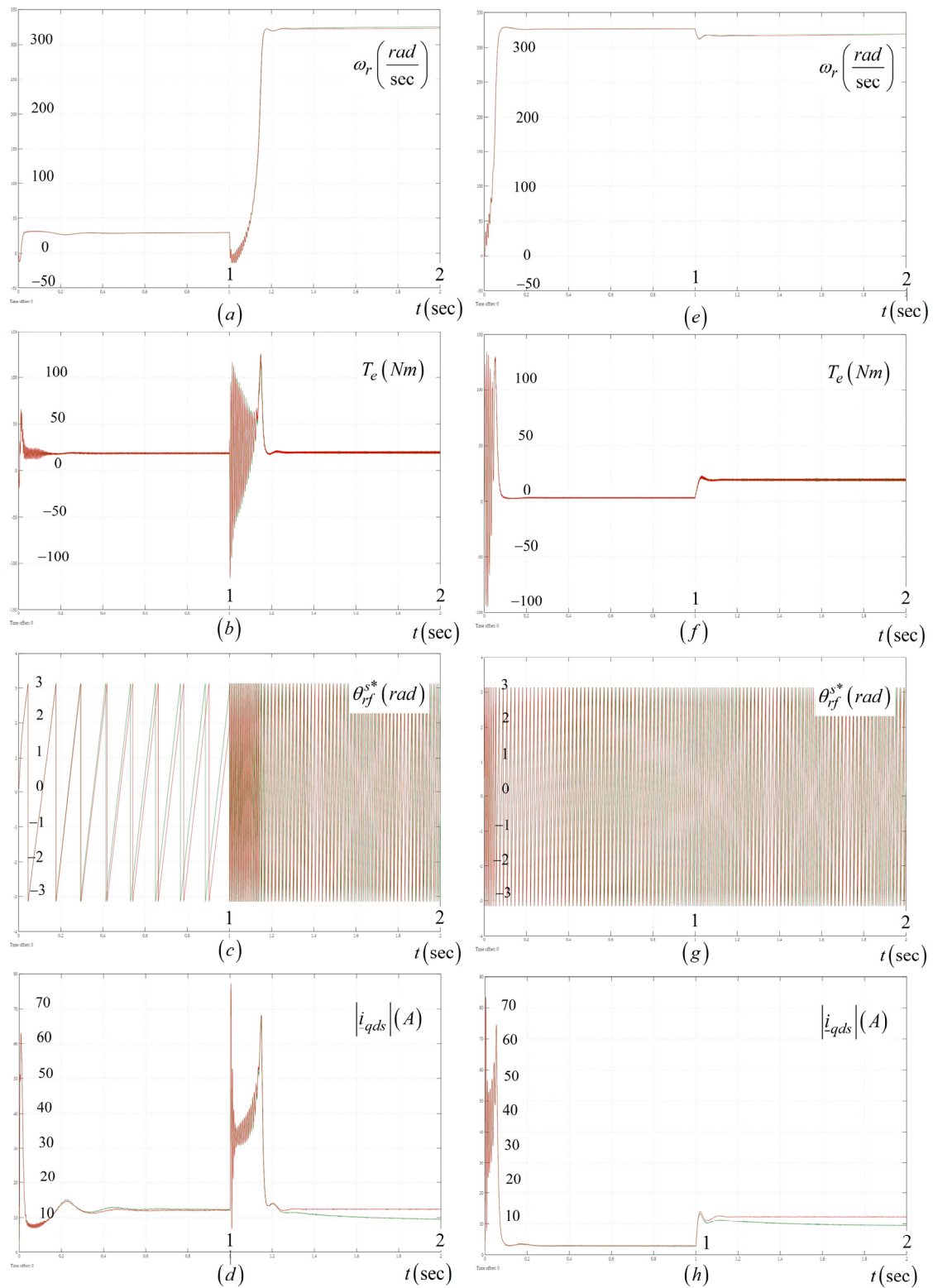


Figure 13. The results of the simulation. Subfigures (a–d) depict the system response under a constant load torque, where a step change in the speed command is applied at $t = 1$ s. Subfigures (e–h) present the system behavior under a constant speed command, with a sudden change in the load torque introduced at $t = 1$ s. Specifically: (a,e) show the rotor speed response; (b,f) illustrate the variation in electromagnetic torque; (c,g) compare the rotor flux angle estimated by two different algorithms; (d,h) represent the magnitude of the stator current phasor.

The simulation results indicate that the proposed method and the conventional indirect FOC exhibit comparable performance in terms of dynamic response. Following step changes in either the speed command or the load torque, both control strategies successfully converge to the new steady-state operating point.

However, under steady-state conditions, the proposed method yields significantly lower stator current magnitudes compared to the conventional FOC approach, indicating improved efficiency. This improvement is particularly evident in Figure 13d,h.

4. Experiment

To validate the proposed approach, both the proposed and conventional indirect FOC methods were implemented on a hardware testbench, as shown in Figure 14. The testbench utilized Texas Instruments' motor control platform (TMDSHVMTRINSPIN), which is based on the TMS320F28335 digital signal processor (DSP). The development environment consisted of Code Composer Studio v5, along with the InstaSPIN library provided by TI (Texas Instruments founded in 1930; Headquartered in Dallas, TX, USA; <https://www.ti.com>).

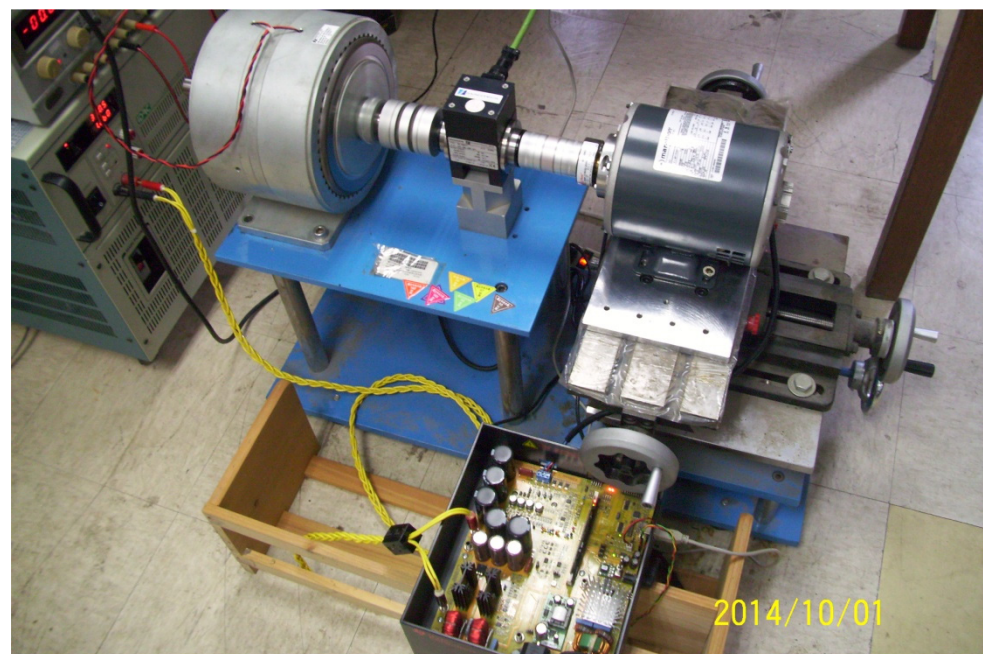


Figure 14. The picture of the testbench.

Table 2 summarizes the specifications of the induction motor used in the experiments. The control program was developed by extending the standard induction motor drive project available in the InstaSPIN library. This baseline project, verified and promoted by TI, served as the foundation for integration. The proposed control algorithm was implemented by embedding the estimation and switching logic modules into the existing control framework.

The experiment was conducted under four combinations of speed commands and load torques: 720 rpm and 1620 rpm for speed, and 0 Nm and 1 Nm for torque. Representative results are shown in Figure 15, which records the phase-A stator current under each condition. The root mean square (RMS) values were calculated under steady-state operation. In the figure, the dashed lines represent the performance of the conventional indirect FOC method, while the solid lines correspond to the proposed method.

Table 2. The ratings and parameters of this real motor.

Quantity	Rated Value
Power	1/4 (hp)
Speed	1800 (rpm)
Voltage	238 (V)
Frequency	60 (Hz)
Number of poles	4
Rotor's resistance	$r_r = 6.11 (\Omega)$
Stator's resistance	$r_s = 11.05 (\Omega)$
Inductance of flux leakage	$L_{ls} = L_{lr} = 0.022484 (H)$
Mutual inductance	$L_m = 0.293939 (H)$

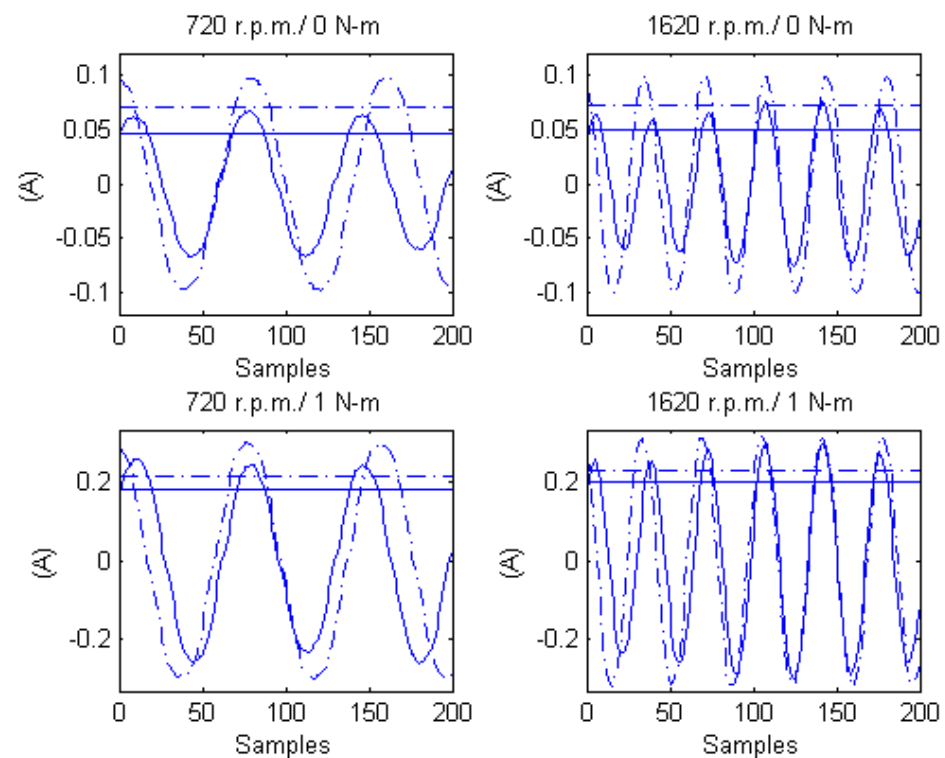
**Figure 15.** The waveforms of phase A current in the steady state and their rms values under different conditions.

Figure 16 summarizes several RMS values of the phase-A stator current measured under different load torque conditions at two speed commands: 720 rpm and 1620 rpm. The experiment utilized the MAGTROL HB-3500 dynamometer, which generates output torque in response to an input current command. Therefore, the horizontal axis of the figure represents the current input to the dynamometer, which serves as a proxy for load torque.

Additionally, since the rated current of the tested motor is 1.2 A, this value is defined as 1 per unit (pu) in the experimental environment. Accordingly, the vertical axis displays the RMS stator current in pu, as observed from the DSP development environment. As in previous figures, dashed lines indicate the performance of the conventional indirect FOC method, while solid lines represent the proposed method.

In summary, compared to the conventional indirect FOC method, the proposed technique consistently results in significantly lower steady-state stator current while maintaining comparable torque and speed responses. The reduction in stator current contributes to decreased power transmission losses, enables the use of lower-rated components, and potentially reduces overall system cost.

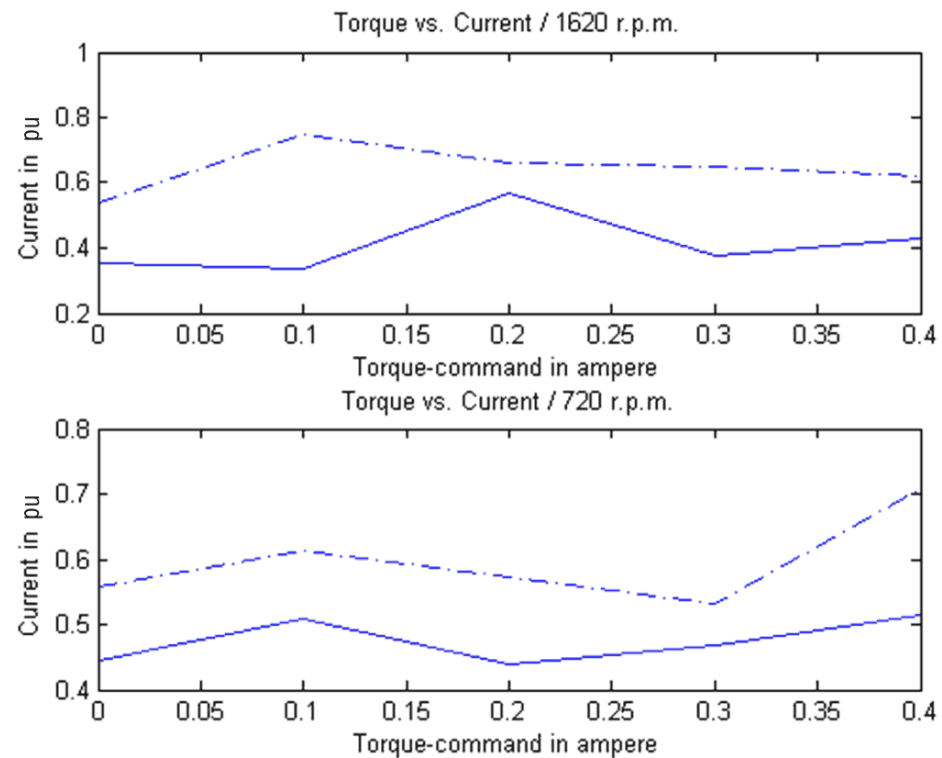


Figure 16. The rms value of phase A current vs. the torque load under different speed commands.

To enhance the comprehensiveness of the manuscript, a comparative summary is presented in Table 3. This table clearly highlights the distinctive features and advantages of the proposed method over other commonly adopted sensorless control strategies for induction motors. Specifically, the comparison encompasses key performance aspects, including robustness to parameter variations, transient response, steady-state accuracy, computational complexity, implementation cost, sensitivity to noise, and practical applicability.

Table 3. Comparison of the proposed method with other common sensorless control strategies for induction motors.

Feature	Proposed Method (Hybrid PLL-Based FOC)	Extended Kalman Filter (EKF)	Sliding Mode Observer (SMO)	Conventional Indirect FOC
Rotor Flux Angle Estimation	PLL-based, pure integration-free	Kalman filter estimation	Observer-based estimation	Integration-based slip estimation
Accuracy (Steady-state)	High (precise PLL alignment)	Moderate-High (depends on model accuracy)	Moderate (affected by chattering)	Moderate (integration drift)
Transient Performance	Good (hybrid switching mechanism)	Good (adaptive estimation)	Good (fast observer response)	Good (established method)
Robustness to Parameter Variations	Moderate (parameter sensitivity exists, can be mitigated by online identification)	Moderate (model-sensitive)	High (robust against uncertainties)	Moderate-Low (sensitive to rotor parameters)
Computational Complexity	Moderate (PLL and simple switching logic)	High (complex computations)	Moderate-High (nonlinear observer)	Low (simple slip calculation)
Implementation Cost	Moderate (standard DSP, no sensors required)	High (advanced DSP required)	Moderate-High (requires robust hardware)	Low (standard DSP sufficient)

Table 3. Cont.

Feature	Proposed Method (Hybrid PLL-Based FOC)	Extended Kalman Filter (EKF)	Sliding Mode Observer (SMO)	Conventional Indirect FOC
Sensitivity to Noise	Moderate (PLL filtering capability)	Moderate-High (Kalman filter requires tuning)	Moderate (observer robust but chattering issue)	Moderate-Low (integration issues)
Practical Implementation and Validation	Validated experimentally and in simulation	Validated mostly in simulation	Experimental validation common	Established practical use
Efficiency Improvement (Stator Current Reduction)	Significant improvement	Moderate (dependent on accurate estimation)	Moderate (efficiency not primary focus)	Moderate (standard approach)
Error Accumulation	Minimal (PLL continuously corrects angle)	Moderate (cumulative estimation errors)	Moderate (observer inaccuracies)	High (integration error accumulates)

5. Conclusions

It is worth emphasizing that the proposed framework is neither a conventional direct field-oriented control (DFOC) nor an indirect field-oriented control (IFOC) scheme. Instead, it combines the advantages of both approaches while avoiding their respective drawbacks. For example, as shown in Figure 3b, the conventional IFOC method calculates the slip frequency ($S\omega_e^*$) from command values and integrates it to obtain the rotor flux angle, which is then fed back into the control loop. This approach can easily diverge from actual system behavior. Similarly, the sensorless DFOC structure in Figure 2b continues to rely on the integration of measured signals to compute θ_{rf}^{s*} . However, continuously integrating inevitably accumulates errors and lacks a correction mechanism.

In contrast, the proposed method introduces a dynamic switching mechanism between transient and steady-state modes, with a phase-locked loop (PLL) gradually aligning the estimated rotor flux angle with its true value during steady-state operation. This hybrid approach improves estimation accuracy while maintaining system stability.

Moreover, since the proposed control strategy achieves convergence within a few seconds, it is particularly well suited for applications that operate primarily at constant speeds. These include air conditioning systems, household appliances, electric vehicles, and other energy-efficient industrial drives. In the context of increasing emphasis on energy efficiency and environmental sustainability, control strategies that reduce power consumption without requiring hardware modifications are of significant value. Compared to the conventional indirect FOC method, the proposed approach demonstrates this potential by achieving higher steady-state efficiency.

One potential limitation of the proposed method lies in its sensitivity to variations in motor parameters. However, this challenge can be addressed through well-established system identification techniques [23–25]. For example, in [26], an adaptive control framework was developed by combining observers for unmeasured states, such as rotor flux and electromagnetic torque, with estimators for time-varying parameters, including rotor resistance and load torque. Rather than sacrificing system performance to mitigate parameter sensitivity, online identification methods can be employed to ensure accurate estimation.

At the very least, the proposed method provides a clear framework and tunable structure for rotor flux angle estimation. Simulation results confirm that, when accurate system parameters, such as those obtained via online identification, are available, the proposed approach remains both feasible and effective.

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Nomenclature

All the symbols used in this paper are listed below.

L_r, L_s	Rotor and stator inductance
r_r, r_s	Rotor and stator resistance
L_m	Mutual inductance between rotor and stator
P	The number of poles
λ	The magnetic flux
i	The current
v	The voltage
x^s	The d axis of the coordinate system aligned with the a -phase windings of the stator
x^{rf}	The d axis of the coordinate system aligned with the rotor flux
x_{xs}	The quantity belonging to the stator
x_{xr}	The quantity belonging to the rotor
x_{rf}	The quantity belonging to the rotor flux
x_{qx}	The component on the q axis
x_{dx}	The component on the d axis
$\underline{x}$	The quantity that is a vector or phasor

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